INFRASTRUCTURE AS CODE

Remote Terraform State Management

A production-grade guide to designing and implementing secure, collaborative, and disaster-resilient Terraform state backends on AWS.

S3 Backend | DynamoDB Locking | KMS Encryption | GitHub OIDC | CloudTrail Auditing | CloudWatch Alarms | Cross-Region DR

Production-Grade Project | AWS | Terraform >= 1.0

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SECTION 01

What is Terraform State?

Terraform state is a JSON file that maps your configuration code to real-world resources, stores metadata about resource dependencies, and caches attribute values to improve performance. Every time you run `terraform apply`, Terraform compares this state file against your desired configuration and actual AWS resources to calculate what changes need to be made.

The `terraform.tfstate` file is the heart of your Terraform project. It maintains a mapping between the infrastructure code written in your `.tf` files and the actual resources deployed in your cloud account — including resource IDs, attributes, dependencies, and more.

State File Structure

```
{
    "version": 4,
    "terraform_version": "1.15.0",
    "serial": 42,
    "lineage": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "outputs": {
        "vpc_id": {
            "value": "vpc-0abc123def456",
            "type": "string"
        }
    },
    "resources": [
        {
            "mode": "managed",
            "type": "aws_vpc",
            "name": "main",
            "provider": "provider['registry.terraform.io/hashicorp/aws']",
            "instances": [
                {
                    "attributes": {
                        "id": "vpc-0abc123def456",
                        "cidr_block": "10.0.0.0/16",
                        "enable_dns_hostnames": true
                    },
                    "tags": {
                        "Name": "main-vpc",
                        "Environment": "production"
                    }
                }
            ]
        }
    ]
}
```

Critical State File Components

Component	Purpose	Example

version	State file format version	4 (current)
terraform_version	Terraform version that last modified state	1.15.0
serial	Increments with each state change	42
lineage	Unique ID for state file lifecycle	UUID
outputs	Output values from configuration	VPC ID, ALB DNS
resources	Tracked infrastructure resources	EC2, RDS, S3
dependencies	Resource relationship graph	VPC → Subnet

SECTION 02

Problems with Local State

While the `terraform.tfstate` file plays a critical role in managing your infrastructure, storing it locally comes with serious drawbacks — especially in team environments or production setups.

1. Sensitive Information Exposure

The state file often contains sensitive data such as:

- API keys and authentication tokens
- Database passwords and connection strings
- Resource metadata and configuration values

If this file is stored locally and committed to Git, it can lead to security risks and potential breaches.

2. No Single Source of Truth

When two developers maintain different local `terraform.tfstate` files, conflicting updates, resource mismatches, and state corruption become inevitable. The result is chaos and an unreliable infrastructure state.

3. Lack of Collaboration and Control

Local state lacks essential team features:

- **State Locking** — prevents multiple users from making concurrent changes
- **Versioning** — tracks history of infrastructure changes
- **Secure Sharing** — centralized access for teams
- **High Availability** — no risk of losing state due to a local machine failure

Production Warning: Without disaster recovery for Terraform state, running `terraform plan` will show every resource as "new" — manual cleanup of orphaned resources becomes necessary and a regional outage makes your state completely inaccessible.

SECTION 03

Why Remote State is Recommended

Remote backends address every local-state limitation. Supported backends include:

- Amazon S3 (with DynamoDB for state locking)
- Azure Blob Storage
- Google Cloud Storage
- HashiCorp Cloud Platform (HCP) Terraform

Limitation	Local State	Remote State
Team Collaboration	Members overwrite each other	Shared access from a single source
Concurrent Applies	Corrupts state	DynamoDB locking prevents corruption
Backup & Recovery	Laptop crash loses mapping	S3 versioning + cross-region replication
Sensitive Data	Plaintext on local disk	KMS-encrypted at rest
Audit Trail	None	CloudTrail + CloudWatch logging

Learning Objectives

By completing this project, you will learn:

- Managing Terraform State: Local vs Remote
- Configuring Backend Resource Creation
- Bootstrap — Creating State Infrastructure
- State Locking with DynamoDB
- State File Security and Access Control
- Monitoring S3 Bucket and Lock Table
- Deploy State Infrastructure
- State File Backup and Recovery — Automated Backup Strategy

- Migrating from Local to Remote State

SECTION 04

Architecture Overview

This project implements a secure, keyless CI/CD pipeline for deploying Terraform from GitHub Actions to AWS, using OpenID Connect (OIDC) for authentication — eliminating the need for static AWS credentials.

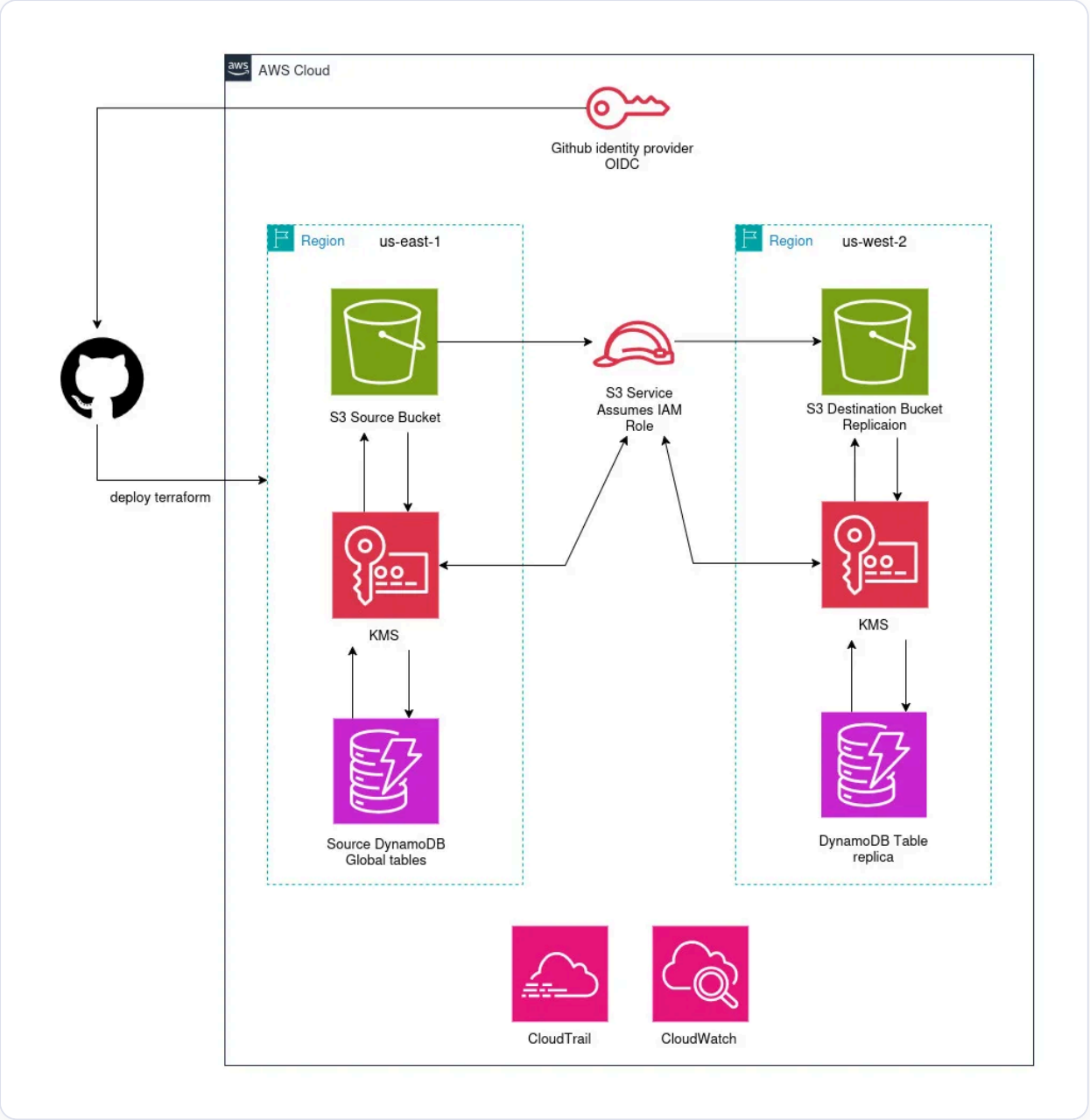


Figure 1 — Detailed component view of the state management architecture

Architecture Components

- **GitHub** — Source of truth; Terraform code lives here and GitHub Actions triggers deployments.
- **GitHub Identity Provider (OIDC)** — Instead of storing long-lived AWS credentials, GitHub authenticates via OpenID Connect with short-lived JWT tokens. No static keys needed.
- **S3 Source Bucket (us-east-1)** — Stores the `terraform.tfstate` remotely in the primary region, enabling team collaboration and state history versioning.
- **S3 Destination Bucket (us-west-2)** — Cross-region replication target for disaster recovery.
- **DynamoDB Global Table** — Handles state locking to prevent concurrent Terraform runs from corrupting shared state. Replicated to the secondary region.
- **KMS Keys** — Customer Managed Keys (CMK) for encrypting S3 state files, CloudTrail logs, and DynamoDB tables in each region.

OIDC Authentication Flow

Step	Action
1	A push/PR triggers a GitHub Actions workflow
2	The runner requests an OIDC token from GitHub
3	AWS validates the token against the registered GitHub identity provider
4	The runner assumes an IAM role (no static credentials)
5	Terraform accesses the S3 backend to read/write state
6	DynamoDB locks the state during the operation

Benefit	Detail
No static credentials	OIDC tokens are short-lived and scoped
Shared state	S3 enables team collaboration
Safe concurrency	DynamoDB prevents state corruption
Auditability	S3 versioning keeps full state history

> **Best Practice:** This OIDC + S3 + DynamoDB pattern is considered the gold standard for managing Terraform in a team or CI/CD environment on AWS.

SECTION 05

Prerequisites & Specification

Prerequisites

- AWS account with appropriate permissions
- AWS CLI configured with access to the target account
- Terraform >= 1.0 installed
- OpenID Connect identity provider registered in Amazon IAM with a trust relationship to your GitHub repository
- IAM Role ARN stored as a GitHub secret named `AWS_ROLE_ARN`

Required AWS Permissions

Your AWS user/role needs full access to: **S3**, **DynamoDB**, **KMS**, **SNS**, **CloudTrail**, and **CloudWatch** (dashboard and metrics).

S3 Bucket Specification

- ✓ Prevent accidental deletion
- ✓ Enable versioning to keep state history
- ✓ Enable KMS encryption at rest
- ✓ Block all public access (critical security)
- ✓ Enable access logging on the state bucket
- ✓ S3 lifecycle policy for cost optimization (STANDARD_IA → GLACIER_IR → DEEP_ARCHIVE)
- ✓ S3 Event notifications to SNS for `ObjectCreated` and `ObjectRemoved`

- Cross-region replication for disaster recovery
- Replicate encrypted objects to secondary region

DynamoDB Specification

- KMS encryption at rest
- Prevent accidental deletion via lifecycle protection
- Global table replication to disaster recovery region
- Point-in-Time Recovery (PITR) enabled

SECTION 06

Implementation — Modules

Project Structure

```
remote-backend-setup/
├── modules/
│   ├── iam/
│   │   ├── bucket_policy.tf      # S3 bucket security policies
│   │   ├── github_actions.tf     # OIDC + IAM role for GitHub Actions
│   │   ├── iam_cloudtrail.tf     # CloudTrail IAM permissions
│   │   └── iam_replication.tf    # S3 replication IAM role
│   ├── kms/                      # KMS key definitions (4 keys, 2 regions)
│   ├── dynamodb/                 # State locking table + global replication
│   ├── s3/                       # State bucket, access logs, replication
│   ├── sns/                      # Alert topics and email subscriptions
│   ├── cloudtrail/               # Audit logging trail
│   └── cloudwatch/               # Monitoring alarms and dashboards
├── main.tf
├── variables.tf
├── outputs.tf
└── .github/workflows/
    ├── backend-deployment.yml    # Deploy state backend via OIDC
    ├── terraform-checkov.yml     # Security scanning
    ├── terraform-tflint.yml      # Linting
    ├── terrascan.yml             # Vulnerability scanning
    ├── tfsec-scanner.yml         # Misconfiguration detection
    ├── terraform-infracost.yml   # Cost estimation
    └── drift-detection.yml       # Infrastructure drift detection
```

MODULE: IAM

BUCKET_POLICY.TF — DEFENSE-IN-DEPTH S3 POLICY

Attaches a JSON policy to the state S3 bucket with four independent security layers:

Statement	What it Blocks/Allows	Why
DenyInsecureTransport	Blocks HTTP requests	Encrypt data in transit
DenyUnencryptedObjectUploads	Blocks non-KMS uploads	Enforce encryption at rest
EnforceMFAForDelete	Blocks deletes without MFA	Protect state from accidental deletion

AllowTerraformRoleAccess	Allows GitHub Actions via OIDC	CI/CD access without static credentials

> **Defense-in-Depth:** Multiple independent security layers ensure that no single misconfiguration can expose your Terraform state.

GITHUB_ACTIONS.TF — OIDC AUTHENTICATION SETUP

Three key resources establish keyless GitHub-to-AWS authentication:

1. **aws_iam_openid_connect_provider** — Registers GitHub as a trusted Identity Provider in AWS IAM
2. **aws_iam_role** — The IAM role that GitHub Actions will assume using OIDC tokens
3. **aws_iam_role_policy_attachment** — Attaches permissions to the role (scope down from AdministratorAccess in production)

IAM_REPLICATION.TF — S3 REPLICATION PERMISSIONS

Creates the IAM role and policy that grants the S3 service permission to replicate objects cross-region. Three permission groups:

- **Group 1** — Read source bucket replication configuration and object list
- **Group 2** — Read each object's content, metadata, tags, and version info
- **Group 3** — Write, delete, and sync tags in the destination (replica) bucket

MODULE: KMS

Creates four AWS KMS Customer Managed Keys (CMK) across two regions. Every key has `enable_key_rotation = true` for security and compliance hygiene.

Key	Region	Purpose
terraform_state	us-east-1 (primary)	Encrypts the Terraform `.tfstate` file in S3
replication_state_key	us-west-2 (secondary)	S3 cross-region replication — KMS key must be local to destination
cloudtrail	us-east-1 (primary)	Encrypts CloudTrail audit logs
dynamodb_replica	us-west-2 (secondary)	Encrypts DynamoDB global table replica — CMKs are region-scoped

MODULE: DYNAMODB

Creates a DynamoDB table that acts as a distributed lock, preventing concurrent Terraform runs from corrupting shared state.

Feature	Configuration	Purpose
Billing Mode	PAY_PER_REQUEST	Auto-scaling, no capacity planning needed
DynamoDB Streams	NEW_AND_OLD_IMAGES	Required for Global Tables cross-region sync
Lock Attribute	LockID (string)	Terraform writes bucket/path/terraform.tfstate as the key
PITR	Enabled	Restore to any second within last 35 days
Lifecycle	prevent_destroy = true	Terraform refuses to destroy this resource
Encryption	Customer CMK via KMS	Full control over key rotation and access
Replica	us-west-2 with own CMK	DynamoDB Global Table for multi-region locking

MODULE: S3

S3-STATE-FILE.TF — SECURE STATE BUCKET

Configures a production-grade state bucket with nine layers of configuration:

1. **Random Suffix** — 8-character suffix for global S3 bucket name uniqueness
2. **Main State Bucket** — Primary bucket storing all `.tfstate` files
3. **Versioning** — Enables state history for rollback on failed applies

4. **KMS Encryption** — SSE with customer-managed key; `bucket_key_enabled = true` reduces KMS API costs
5. **Public Access Block** — All four block settings enabled — state files contain sensitive values
6. **Lifecycle Policy** — Automatic cost optimization via storage tier transitions
7. **Access Logs Bucket** — Separate bucket for S3 access audit logs, auto-expired after 90 days
8. **Access Logging** — Ships logs to the logs bucket under `terraform-state-access-logs/`
9. **SNS Notifications** — Publishes events to SNS on `*.tfstate` creates/deletes

LIFECYCLE COST OPTIMIZATION

Transition	Storage Class	Cost Tier
After `standard_ia` days	STANDARD_IA	Infrequent access — lower cost
After `glacier_ir` days	GLACIER_IR	Archival — very low cost
After `deep_archive` days	DEEP_ARCHIVE	Cheapest available storage

REPLICATION.TF — CROSS-REGION DISASTER RECOVERY

Primary Region (us-east-1)	Secondary Region (us-west-2)
terraform_state bucket KMS encrypted, versioned	state_backup_replica bucket Re-encrypted with regional CMK, STANDARD_IA, versioned

MODULE: SNS

Sets up two SNS topics with email subscriptions for state change notifications:

- **terraform-state-alerts** — General alert topic for CloudWatch alarms
- **s3-event-notification-topic** — Dedicated topic for S3 bucket object events, requiring an explicit resource policy for S3-service publishing

All addresses in `var.email_subscriptions` are subscribed to both topics automatically. Recipients receive an AWS confirmation email before notifications begin.

MODULE: CLOUDTRAIL

Implements complete API-level audit logging for all Terraform state operations.

CLOUDTRAIL CONFIGURATION

- **enable_log_file_validation = true** — Digest files for cryptographic tamper detection
- **include_global_service_events = true** — Captures IAM, STS, and CloudFront events
- **is_multi_region_trail = true** — Single trail covers all AWS regions
- **CloudWatch Logs integration** — Real-time querying and alerting

AUDIT LOG LIFECYCLE

Event	Trigger
Move to Glacier (cheap archive)	After 90 days
Delete old non-current versions	After 30 days
Clean up incomplete uploads	After 7 days
Permanently delete logs	After 365 days (adjust per compliance)

EVENT SELECTORS (WHAT TO WATCH)

- **S3 Data Events** — Every GetObject, PutObject, DeleteObject on the Terraform state bucket; shows who read or modified state and when
- **DynamoDB Data Events** — Every read/write on the lock table; captures lock acquisitions and releases showing when plan/apply operations ran

MODULE: CLOUDWATCH

A comprehensive monitoring and alerting system with four concern areas — all alarms route to the central SNS topic for unified team notification.

1. TERRAFORM STATE FILE MONITORING

- **state_modification** — Fires immediately on any PutObject to state bucket (threshold = 0)
- **state_access** — Triggers on 5+ accesses within 5 minutes (runaway process or unauthorized access indicator)
- **excessive_state_access** — Higher-severity alarm: 100+ accesses in 5 minutes
- **state_backup_stale** — Daily check confirming at least 1 object exists in state bucket

2. DYNAMODB LOCK TABLE MONITORING

- **long_held_lock_alarm** — Checks every 5 minutes over 12 periods (1 hour); if ItemCount > 1 persists, a crashed Terraform process likely left a ghost lock
- **replication_latency** — Fires when DynamoDB Global Table replication exceeds 5 seconds, risking lock inconsistencies

3. S3 REPLICATION LAG

- **replication_lag** — Monitors S3 CRR latency; fires if replication exceeds 15 minutes over two consecutive 5-minute windows

4. SECURITY ALERTS

Filter	Pattern	Alarm Threshold
oidc_failures	AssumeRoleWithWebIdentity with any errorCode	> 5 failures / 5 min
unauthorized_api_calls	UnauthorizedAccess or AccessDenied errors	> 10 calls / 5 min
root_account_usage	userIdentity.type = "Root", not a service event	≥ 1 usage / 5 min

> **Security Note:** The root account alarm is zero-tolerance. Any direct use of the AWS root account should be treated as an immediate red flag in a well-governed environment. The OIDC filter targets GitHub Actions authentication failures specifically.

SECTION 07

Migrating from Local to Remote State

Follow these steps in order to safely migrate from a local state file to the remote S3 backend:

1. **Initialize and Deploy State Infrastructure**

   Run `terraform init`, `terraform plan`, and `terraform apply` in the `remote-backend-setup/` directory.

2. **Save Backend Outputs**

   Export outputs for later use: `terraform output -json > backend-config.json` and `terraform output backend_config > backend-template.tf`

3. **Backup Local State File**

   Copy it locally and optionally upload to S3: `cp terraform.tfstate terraform.tfstate.backup`

4. **Add Backend Configuration (backend.tf)**

   Activate the generated `backend.tf` file pointing to your new S3 bucket and DynamoDB table.

5. **Reinitialize to Migrate**

   Run `terraform init -migrate-state` and confirm when prompted to copy existing state to the new S3 backend.

6 **Verify Migration**

Run `terraform state list`, `terraform show`, and confirm the state file exists in S3 via `aws s3 ls`.

7 **Remove Local State**

After full verification: `rm terraform.tfstate terraform.tfstate.backup`

terraform init -migrate-state

```
$ terraform init -migrate-state
Initializing the backend...

Do you want to copy existing state to the new backend?
Pre-existing state was found while migrating the previous "local" backend to the
newly configured "s3" backend. No existing state was found in the newly
configured "s3" backend. Do you want to copy this state to the new "s3"
backend? Enter a value: yes
Successfully configured the backend "s3"! Terraform will automatically
use this backend unless the backend configuration changes.

Terraform has been successfully initialized!
$ terraform state list
module.kms.aws_kms_key.terraform_state
module.s3.aws_s3_bucket.terraform_state
module.dynamodb.aws_dynamodb_table.terraform_locks
...
```

Generated Backend Configuration

```
terraform {
  backend "s3" {
    bucket         = "ce-dev-terraform-state-5bhkr8k3"
    key            = "dev/infrastructure/terraform.tfstate"
    region         = "us-east-1"
    encrypt        = true
    kms_key_id     = "cd25fbac-3164-4aad-91a7-7cb1e95e3507"
    dynamodb_table = "ce-dev-terraform-locks"
  }
}
```

Useful State Management Commands

```
$ terraform state list
```

```
$ terraform state show aws_instance.web
```

```
$ terraform state pull > state.json
```

```
$ terraform state mv aws_instance.old aws_instance.new
```

```
$ terraform state rm aws_instance.web
```

SECTION 08

Deployment Workflow & Security Scanning

The project ships with a complete GitHub Actions workflow suite covering automated security analysis, cost estimation, and deployment with OIDC.

Workflow File	Purpose
backend-deployment.yml	Deploy Backend Remote Terraform State Management with OIDC
terraform-checkov.yml	Security scanning via Checkov to detect Terraform misconfigurations at repo and PR level
terraform-tflint.yml	TFLint scans `.tf` files for stylistic errors, security problems, and provider-specific issues
terrascan.yml	Scan Terraform code for security vulnerabilities
tfsec-scanner.yml	Catch security misconfigurations before they reach cloud infrastructure
terraform-infracost.yml	Estimate cloud infrastructure costs before deploying — proactive cost management
drift-detection.yml	Scheduled monitoring for infrastructure drift; catches unauthorized changes and alerts the team early

Teardown

To delete all created resources:

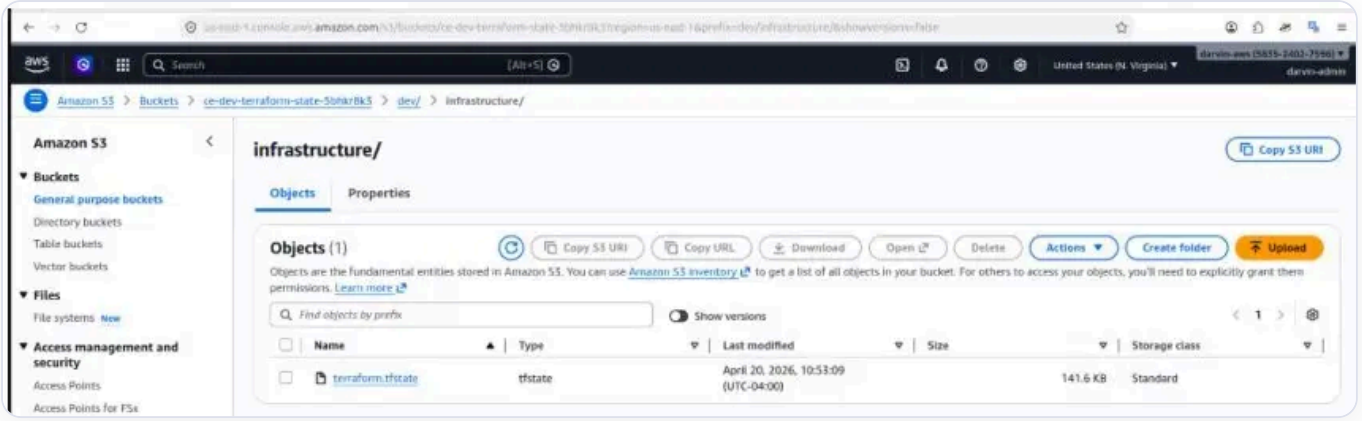
```
$ terraform destroy -var-file="dev-terraform.tfvars" -auto-approve
```

SECTION 09

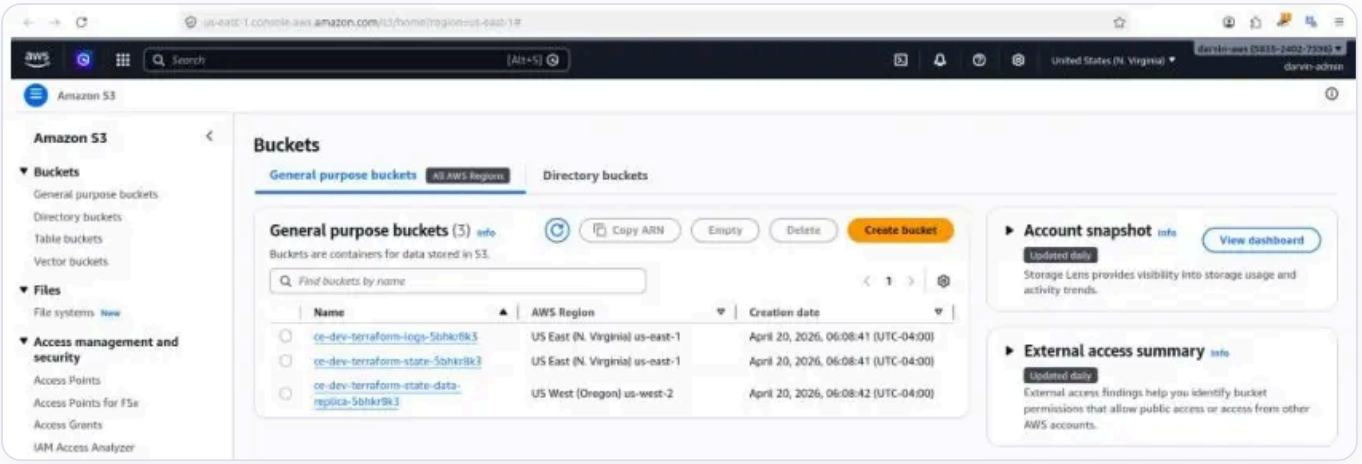
Verification — AWS Console Screenshots

After successful deployment, the following resources are visible in the AWS Management Console:

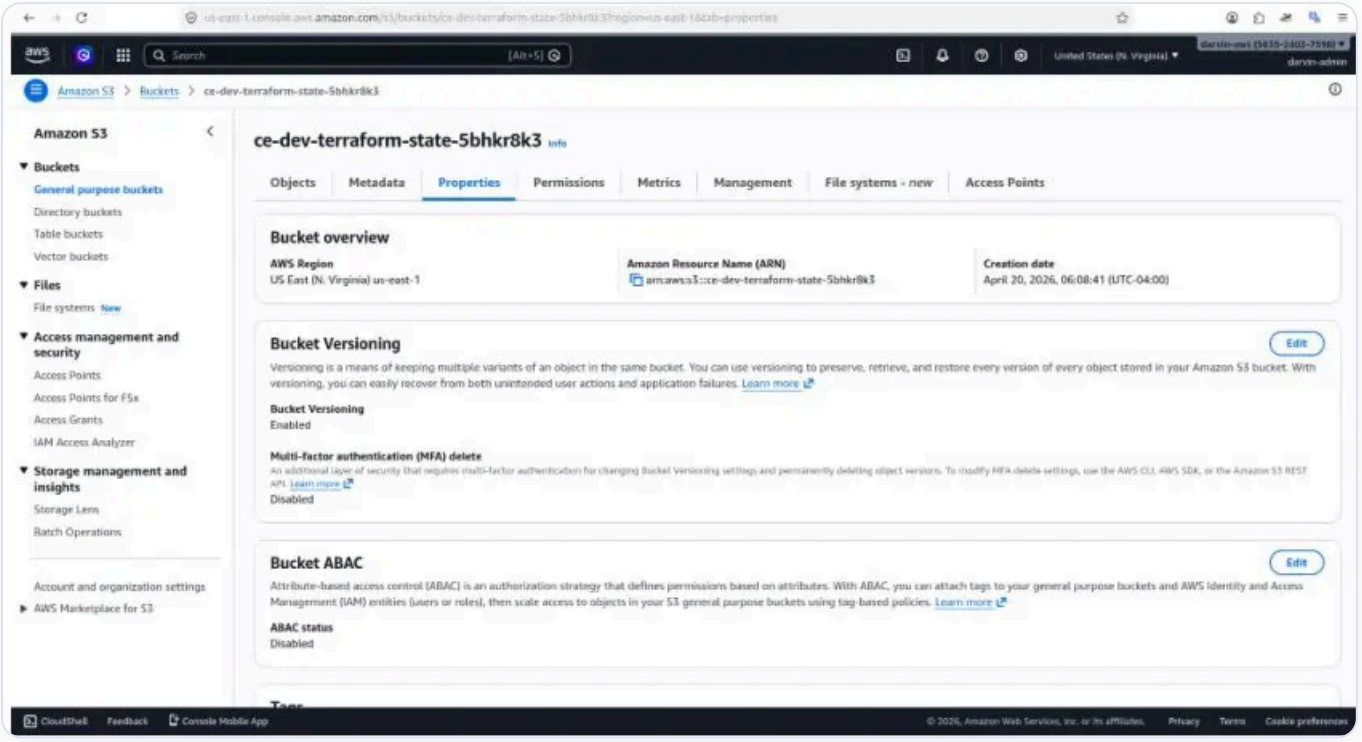
State Migration & S3 Buckets



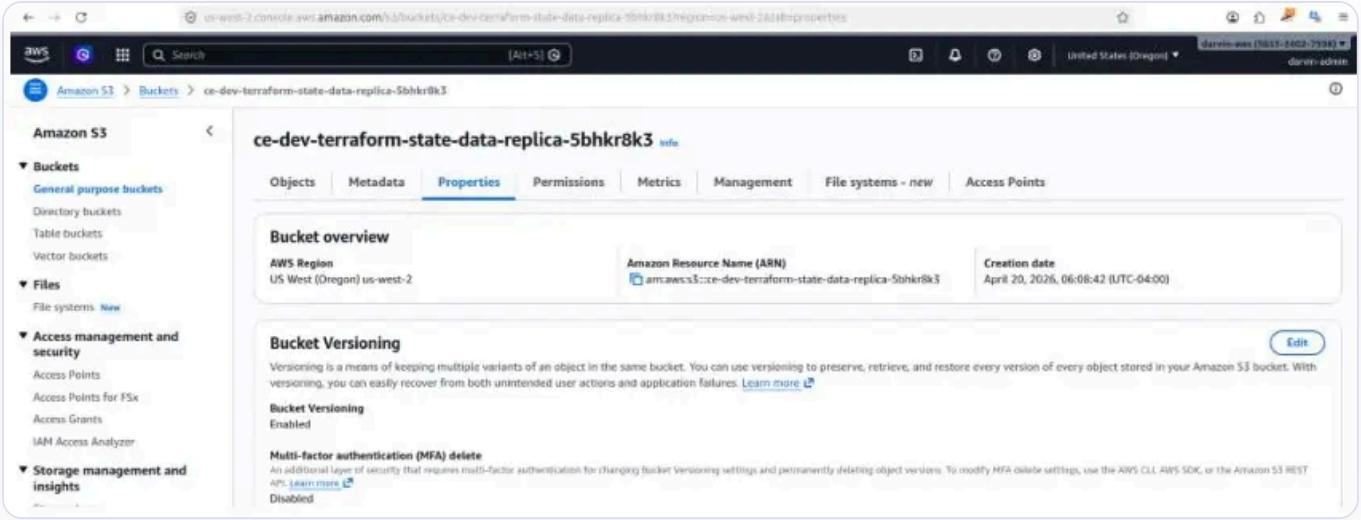
Terraform state migration confirmation



S3 buckets created

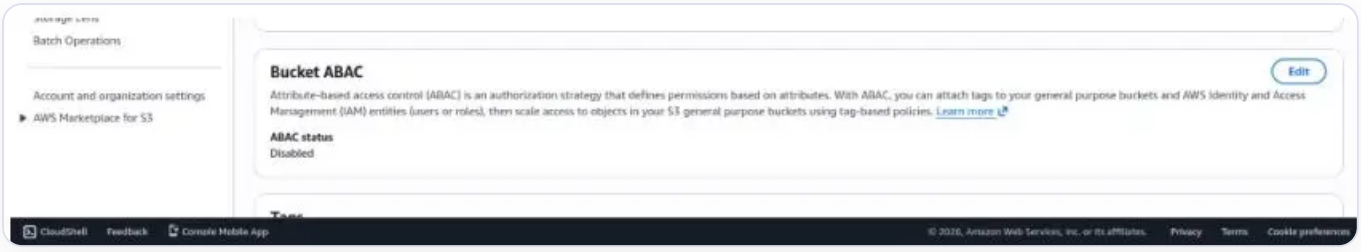


S3 bucket properties — primary state bucket

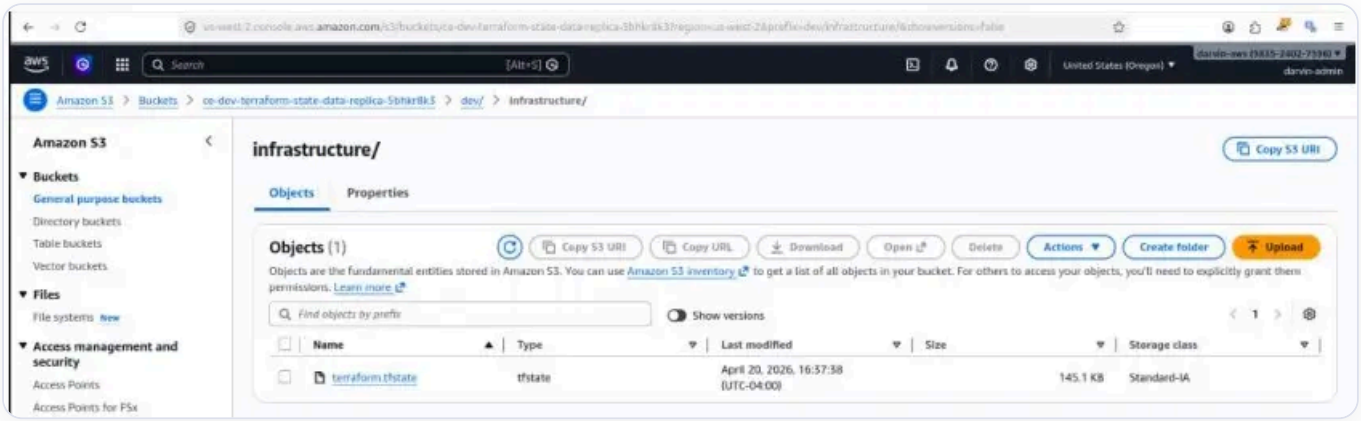


S3 bucket properties — replication bucket

Cross-Region Replication

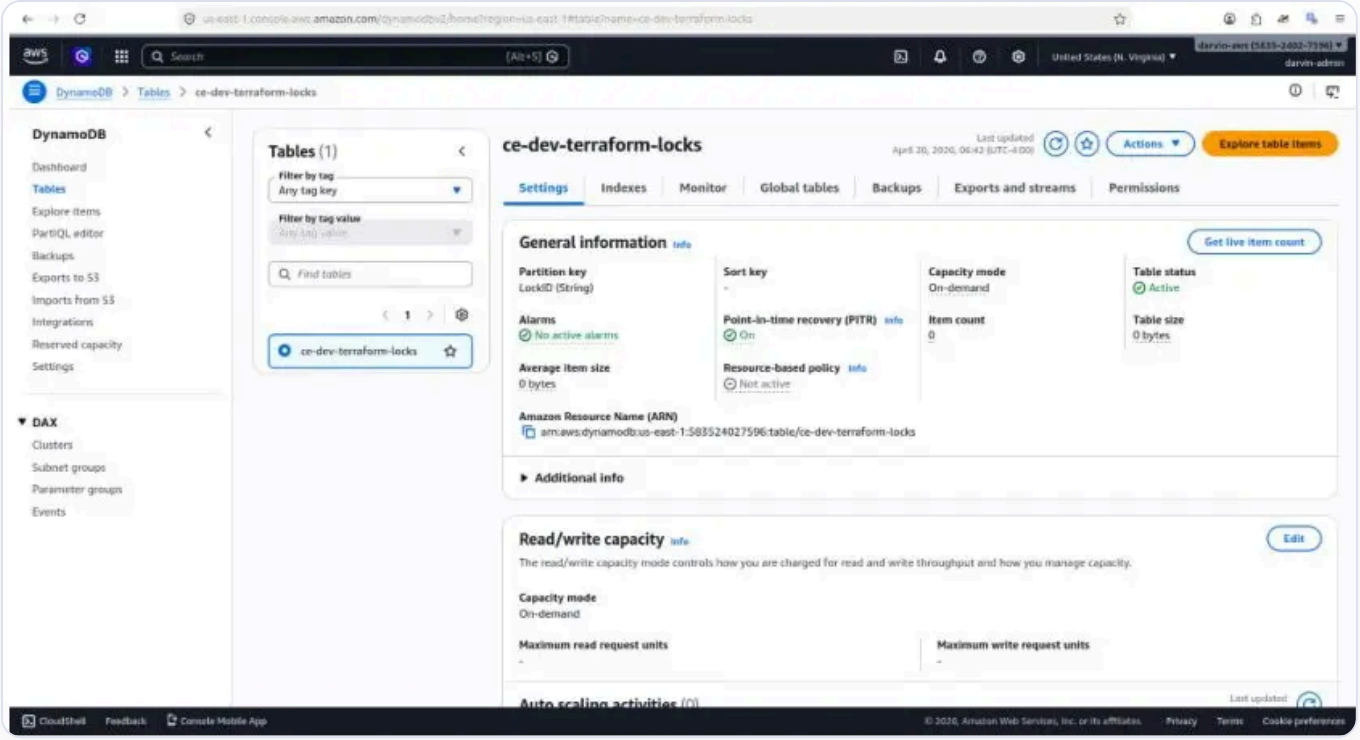


State replication details

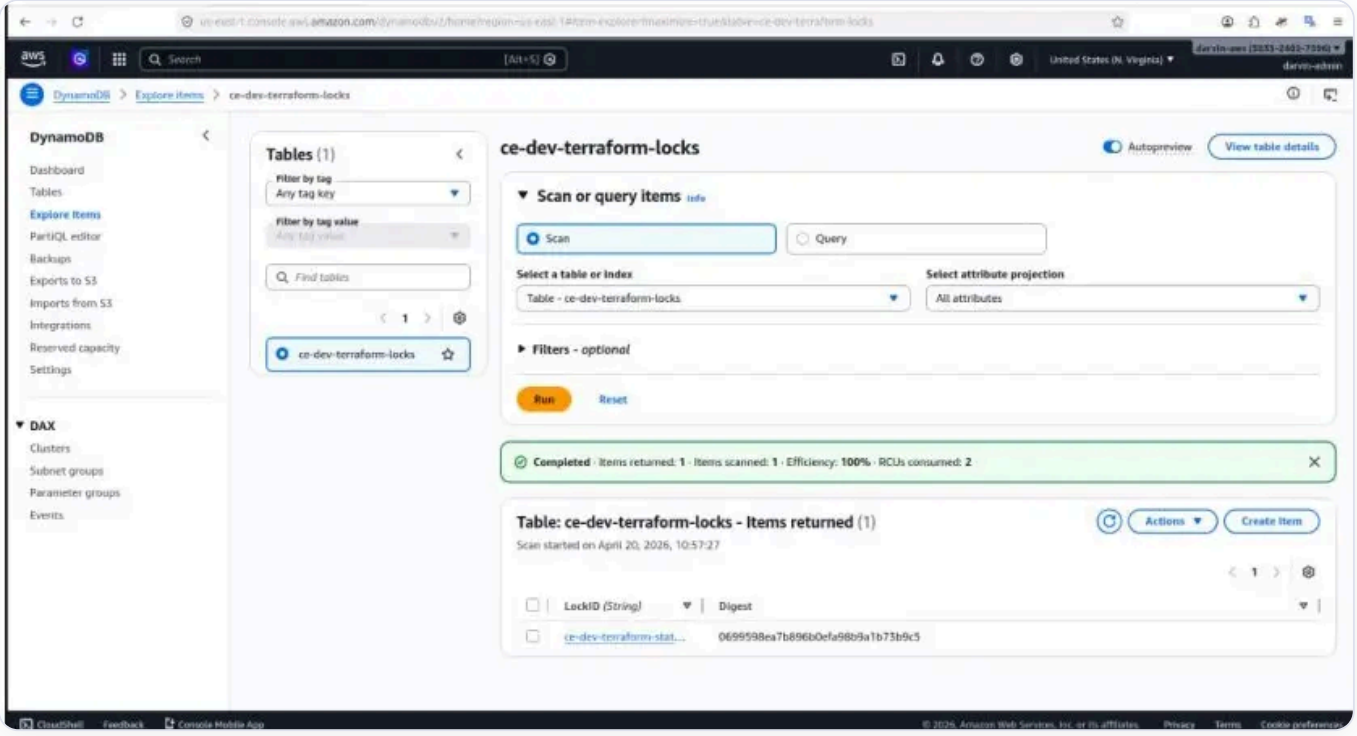


Terraform state replicate in us-west-2

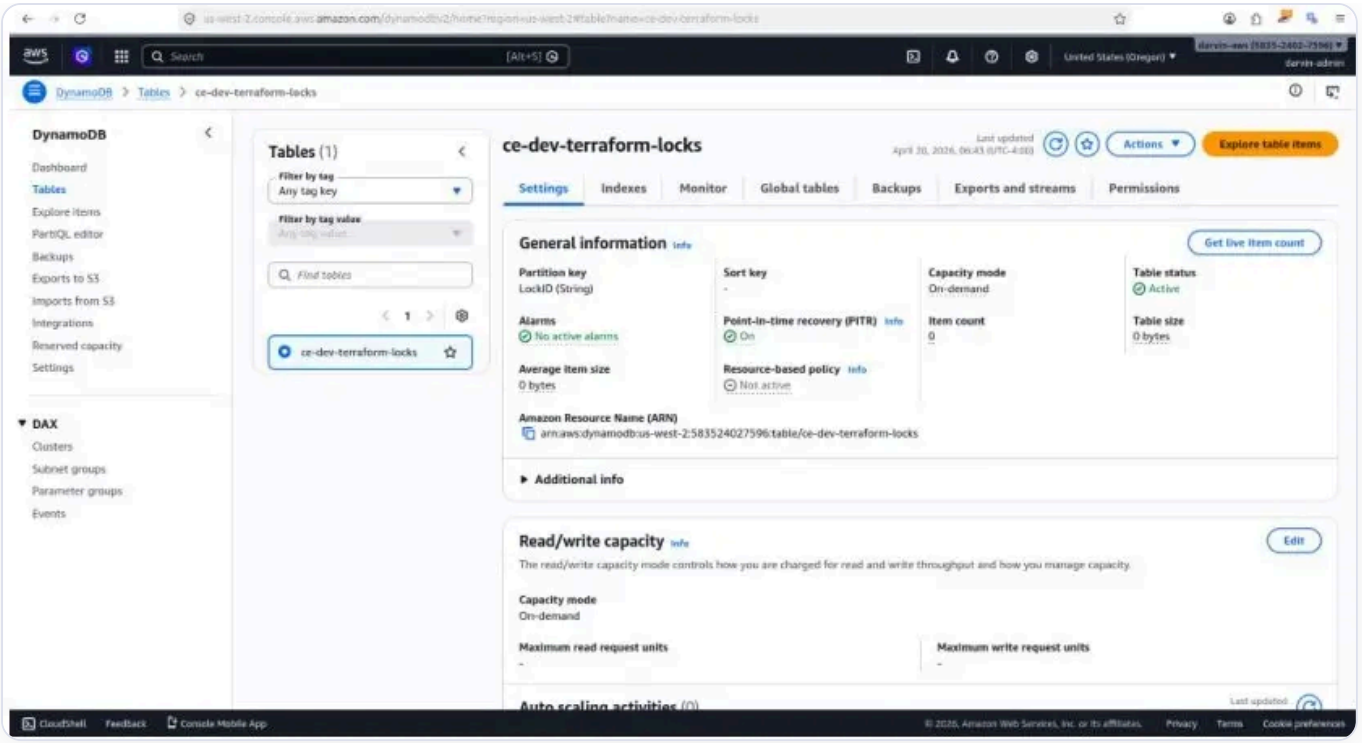
DynamoDB State Locking



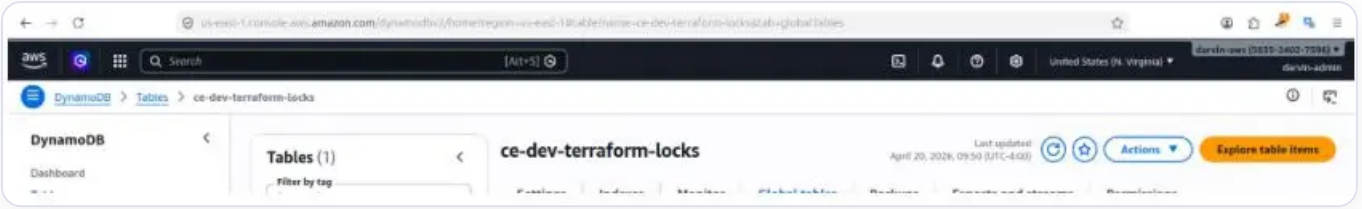
DynamoDB lock table — primary region



DynamoDB table items (lock records)

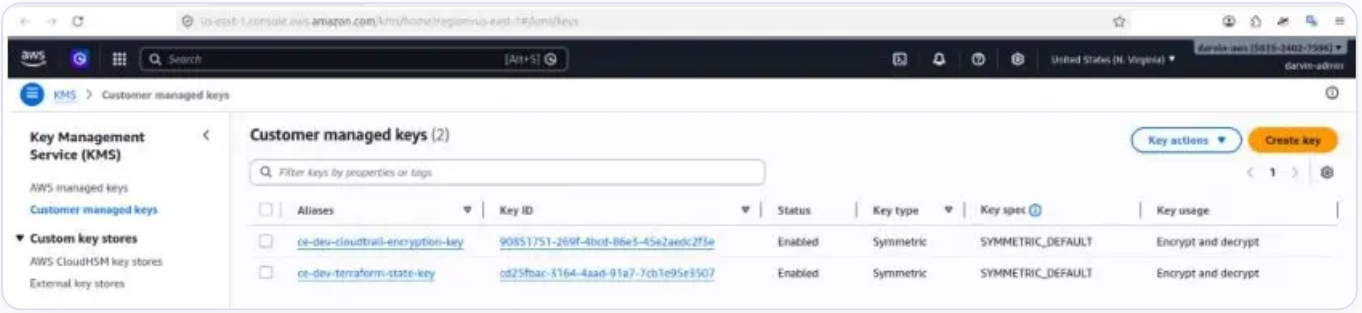


DynamoDB lock table — secondary region

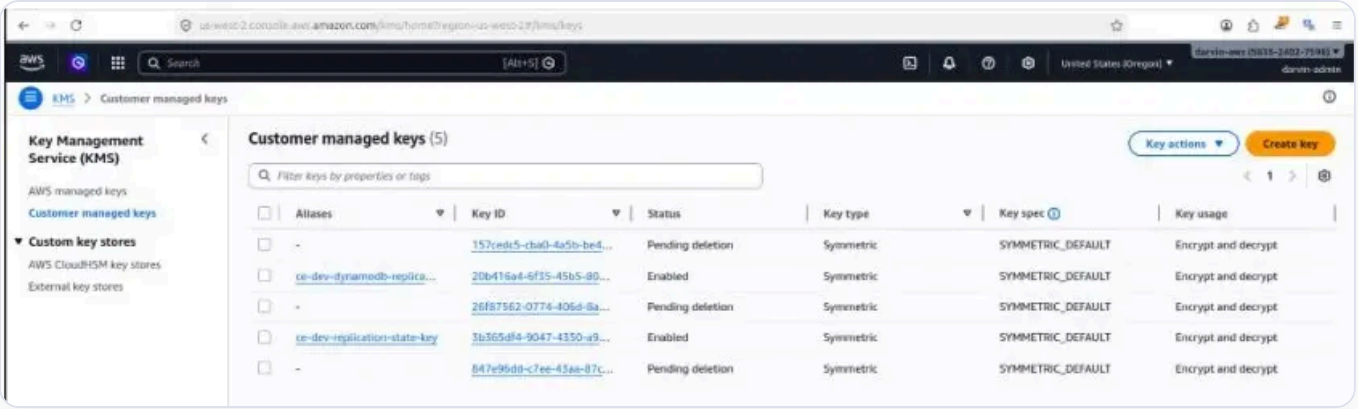


DynamoDB Global Tables replicas

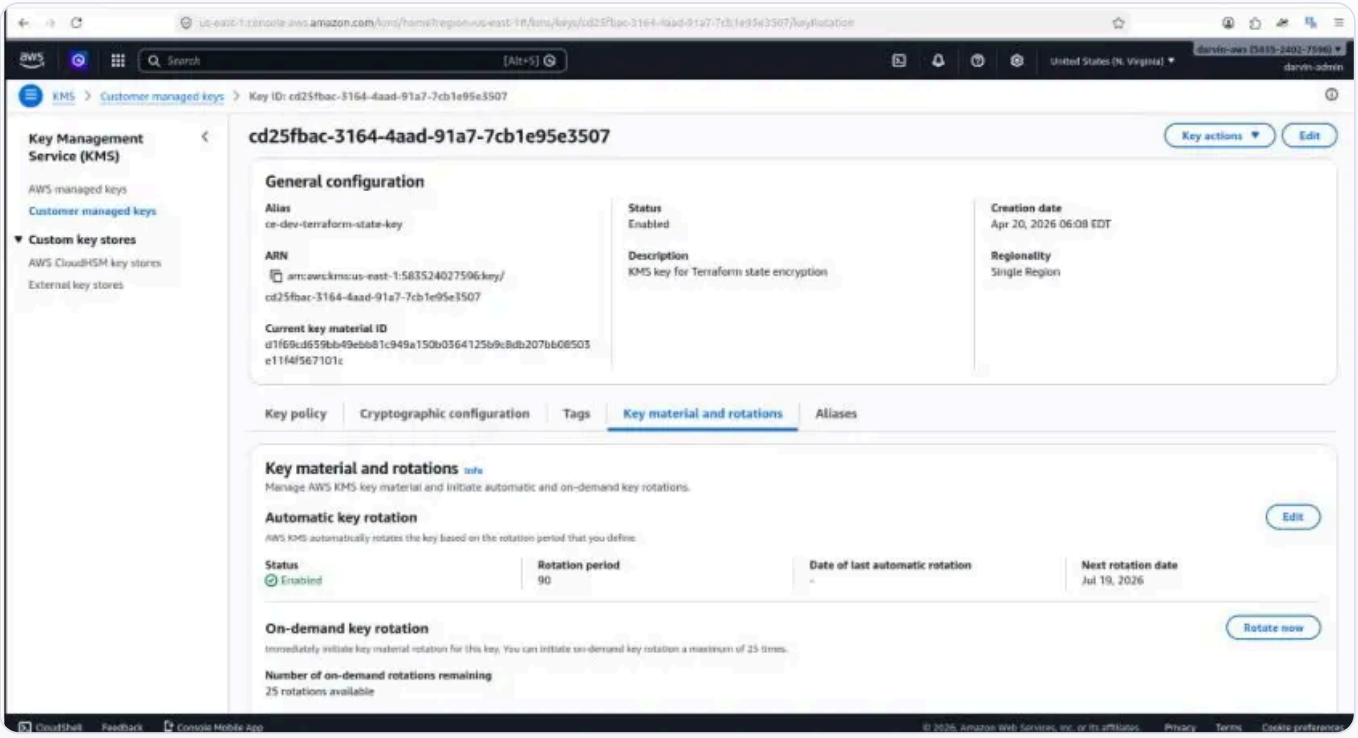
KMS Encryption Keys



KMS key — primary region (us-east-1)

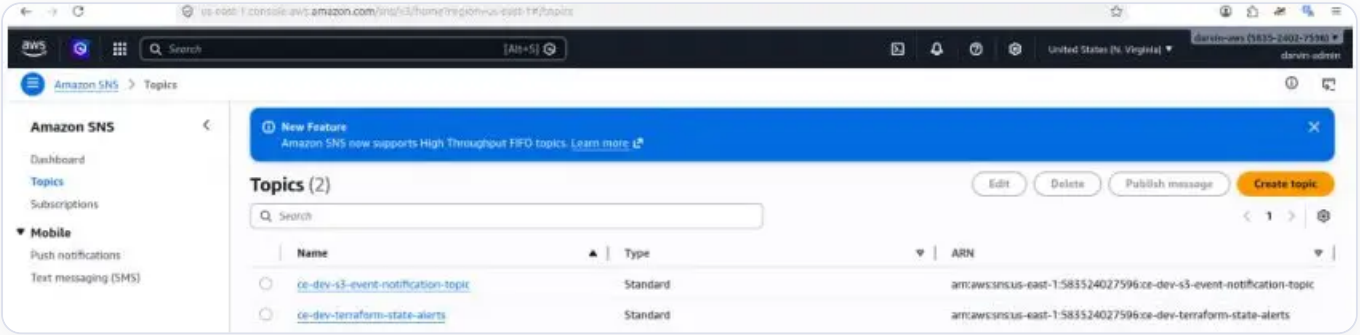


KMS key — secondary region (us-west-2)

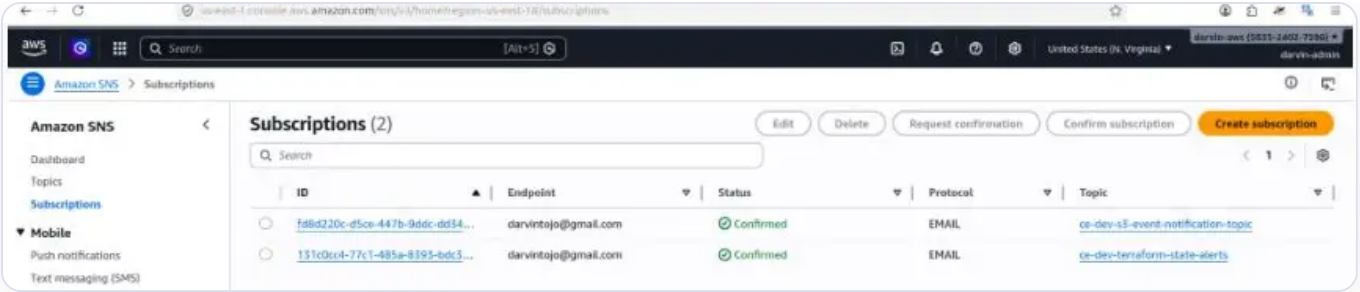


KMS key automatic rotation configuration

SNS Topics & Subscriptions

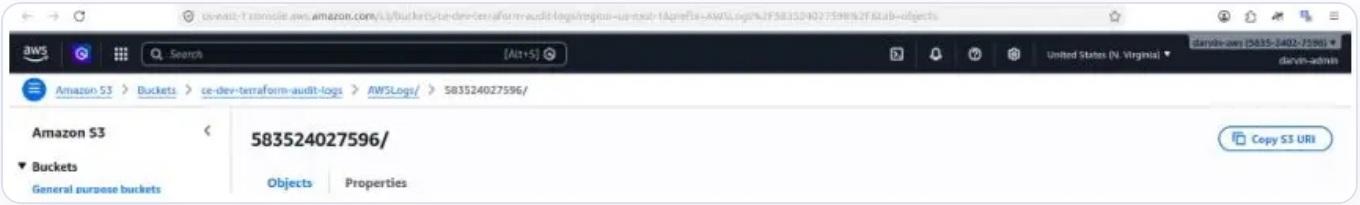


SNS topics created

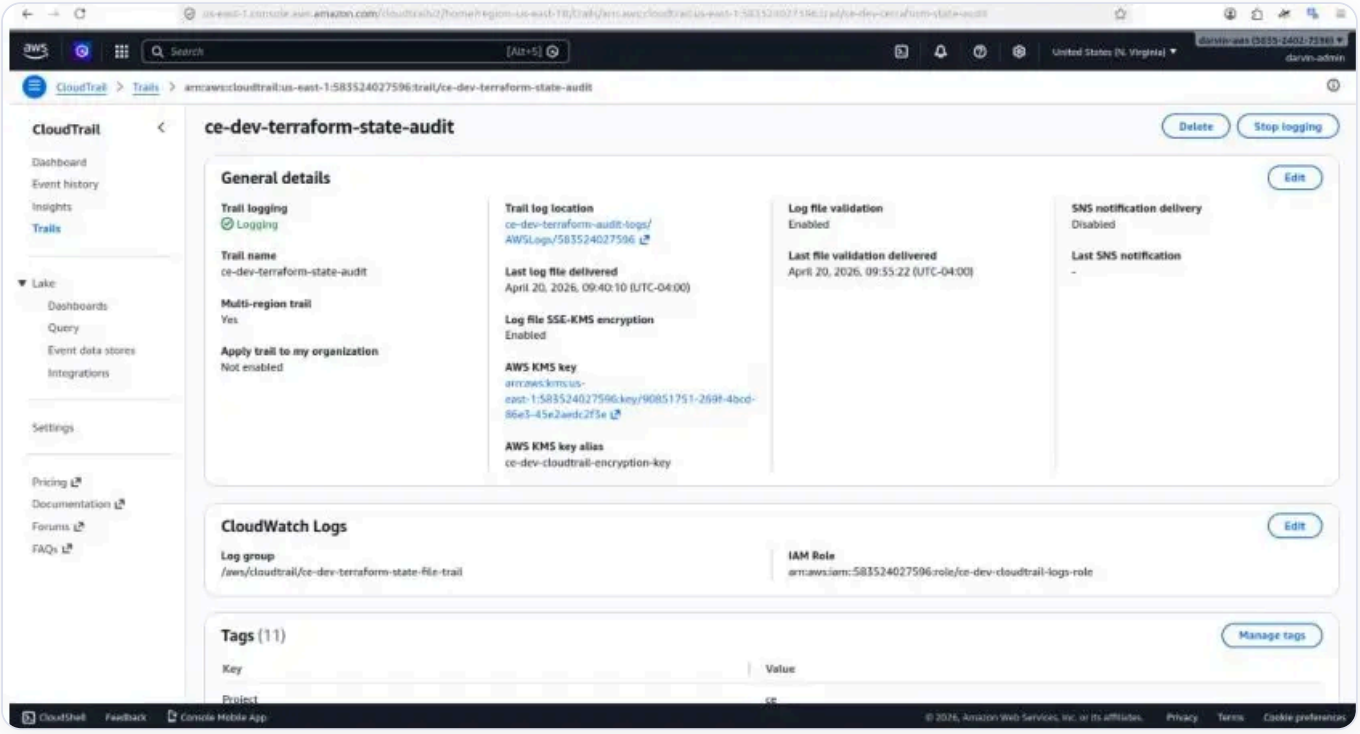


SNS email subscriptions

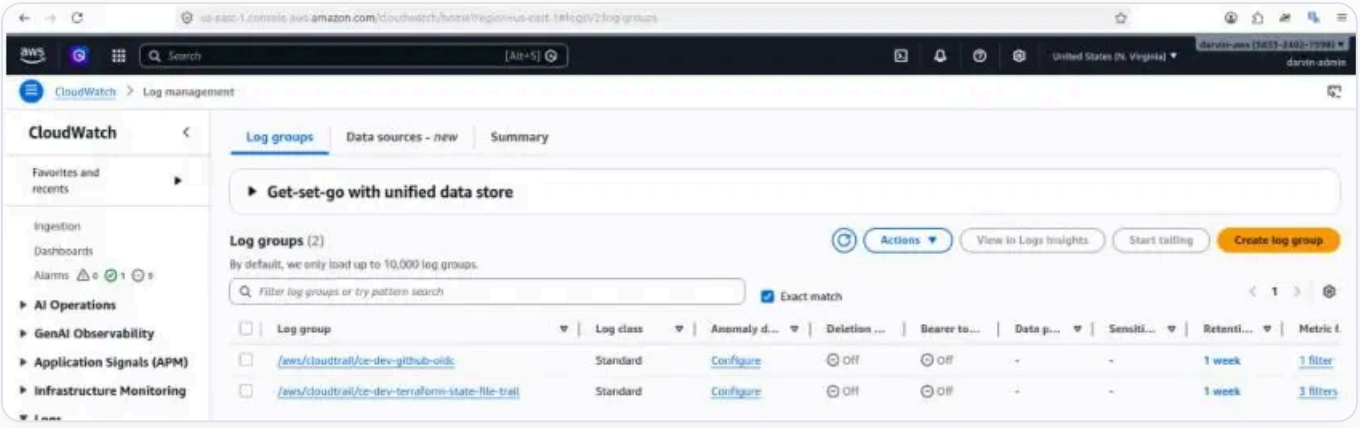
CloudTrail & CloudWatch



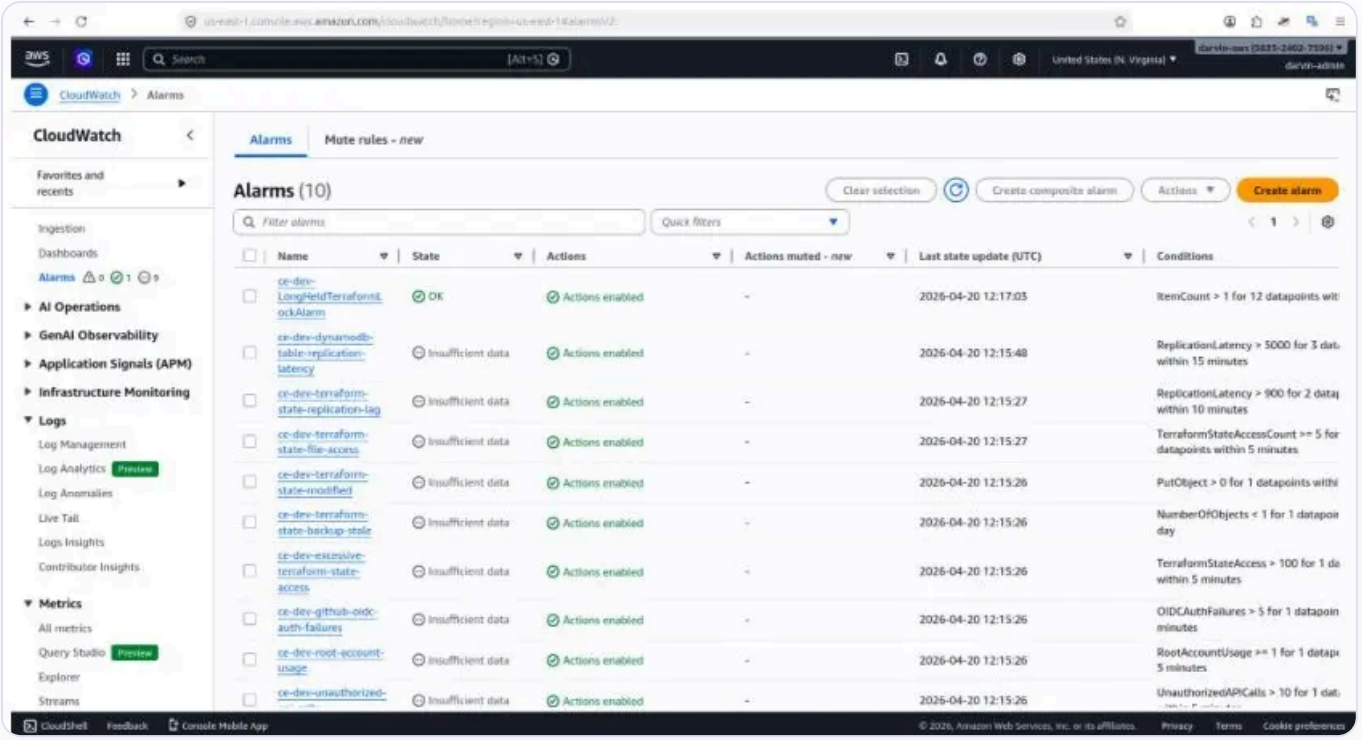
CloudTrail S3 audit bucket



CloudTrail trail — general details



CloudWatch Log Groups for CloudTrail logs



CloudWatch alarms — full list

SECTION 10

Best Practices

Always Encrypt State at Rest

Use KMS CMKs for full control over key rotation and access policies.

Restrict with IAM Policies

Control who can read and write state. Use least-privilege principles.

Enable S3 Versioning

First line of defense against accidental deletion and corruption. Non-negotiable for production.

Cross-Region Replication

Keep a copy of state in a different geographic region to survive regional outages.

Monitor with CloudWatch

Enable metric alarms and alerts; catch issues before they become outages.

Audit Access with CloudTrail

Log all S3 and DynamoDB data events to answer "who changed what and when."

Test Your Restore Process

A backup you have never tested is a backup you cannot trust. Run DR drills regularly.

Keep Backups 30+ Days

Some issues take time to surface. Never store backups in the same location as primary state.

Replicate Lock Tables

Use DynamoDB Global Tables so locking works correctly during regional failover.

Document Failover Procedure

Store it somewhere accessible during an outage — not only in the primary region.

Block All Public Access

State files contain passwords, keys, and ARNs. Never expose state publicly under any circumstance.

Monitor Replication Lag

Stale replicas mean data loss during failover. Alert on lag exceeding acceptable thresholds.

What to Audit

- **Who** made the change (user identity, CI/CD pipeline, service account)
- **When** the change happened (timestamp)
- **What** changed (resources added, modified, or removed)
- **How** the change was made (manual apply, CI/CD, state push)
- **Why** the change was made (PR, ticket, change request linkage)

SECTION 11

Conclusion

Remote state is foundational for production Terraform. The combination of **S3** for storage, **KMS encryption** for security, **versioning** for recovery, and **DynamoDB** for locking gives you a solid foundation for team collaboration.

With this configuration you have a production-ready state management system that handles locking, versioning, encryption, cross-region disaster recovery, real-time alerting, and complete audit logging. Treat state files as sensitive data — control access through IAM, use versioning when you need to troubleshoot or roll back changes, and test your disaster recovery procedures regularly.

> **Production-Ready Checklist:** S3 backend with versioning and KMS encryption ✓ DynamoDB state locking with global table replication ✓ GitHub OIDC for keyless CI/CD authentication ✓ CloudTrail + CloudWatch for auditing and alerting ✓ Cross-region DR with automated replication ✓